

6.3 Groups affected by the digital divide

Here are some examples of ways in which we can compare the digital divide:

- people in different age groups
- people in cities and people in rural areas
- people with differing levels of education
- people in different social-economic groups
- people with accessibility barriers
- people in more and less industrially developed nations.

People in different age groups

Technology is developing at great speed. New technology is released every year. For example, mobile phone companies may release new mobile phones every two years. People born in the 1960s grew up with very limited information technology. Business and personal communications relied on landline telephones (connected to wires and cables). Letters were typed or handwritten. Letters were sent by post or fax machine (see Figure 6.4, using a phone line) rather than using email. Groceries, clothing and other goods were bought at the shops.



Figure 6.4: A fax machine is used to copy and transmit a document down a phone line. The receiving fax machine prints out a copy of the document.

DISCUSSION POINT

Have you ever seen or used a fax machine? Do you think you would know how to use it? Would you just guess? Or would you want to see the user guide?

People born in the 1980s saw new technology emerging. This included simple mobile telephones, the ability to send short text messages, basic low-speed access to the internet using telephone lines, simple websites and downloading items from file storage.

By the beginning of the 21st century, wireless technology and almost instant communication became available. For example, smart phones, using wireless technology, allow instant access to online banking and shopping, the ability to book healthcare appointments, social media, messaging services, email and streaming services.

Many people today have grown up using this technology. They have been taught the IT skills required to use new technology in school or college. Younger people are better able to learn new skills as new features are introduced. On the other hand, some older people may find it hard to learn the new skills needed to use digital devices and the new technology. This is because the technology works in a very different way from the technology that they grew up with.

A person's ability to learn about and use technology effectively is called **digital literacy**. A lack of digital literacy may prevent a person from benefiting from online services. Lack of digital literacy can affect people from any age group. For example, an older person may lack digital literacy because they live in a rural area and do not have access to a local community college to learn about new devices. Without digital literacy, they will lack skills to access digital services or communicate with people effectively using digital devices. A younger person may lack digital literacy because they cannot afford technology and are less confident using it.

The term digital divide describes the difference between people who have and use technology and those who may have difficulty accessing and using new technologies.

ACTIVITY 6.01

Choose an activity that you use a digital device for, such as playing an online game, or creating and sending a short video.

Create a short video or presentation that explains how to do this activity on your digital device.

The video or presentation should teach someone who has never used this technology what to do.

Real-life example

Mohammed is an independent 75-year-old living in his own home. He has a mobile phone so that he can contact his daughter who lives in another town. He doesn't know how to use the mobile phone's other features. He only knows how to phone his daughter.

Mohammed has regular hospital appointments. He used to receive a letter from the hospital reminding him of his appointments, but the hospital no longer does this. Instead, it sends Mohammed text messages, reminding him of the date and time of his appointment, and asking him to confirm by text if he will attend. Mohammed doesn't

know how to send a text message. He knows his daughter is very busy and doesn't want to phone and ask her what to do. Instead, he worries that he will lose his hospital appointment (see Figure 6.5).

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Figure 6.5: People who lack digital literacy skills may be at a disadvantage.

UNPLUGGED ACTIVITY 6.02

Design a poster to show how technology in a shopping centre may affect people with low digital literacy. Include:

- the technology used
- the challenge faced by someone with low digital literacy
- the benefits of this technology to the shopping centre or store.

Reducing the digital divide between different age groups

One of the biggest barriers to people using technology is fear of new things. They may be more comfortable doing things in ways that are familiar to them. For example, they may prefer to visit different shops to look for an item that is difficult to find, rather than search the internet to find the item they need. A person may believe that they are not clever enough to learn how to use digital devices. Quite often, older people feel more scared learning about new technology. Technology changes so fast that they feel they 'cannot keep up' with all the changes.

DISCUSSION POINT

Imagine you have been asleep for 100 years. You wake up and technology has developed. The following technologies have been developed:

- teleporters
- fully automatic robots to help you in the home
- flying cars

- machines that make all your meals
- headsets that read your thoughts to operate things such as the television, your car and write emails.

Discuss with a classmate:

- How would you feel?
- What challenges would you experience?
- What things would you need to help you adapt to the new technology?
- How does this relate to how older people or those with low technology skills feel in today's world?

People often choose to learn new things when they see the benefits of learning the new skill. People are more likely to learn to use new technology if they see that the benefits will be greater than the challenge of learning the new skill. For example, learning how to use social media may allow people to communicate with family and friends and prevent them from being socially isolated and combat loneliness. Many countries support older people to help them appreciate the advantages of technology and teach them how to use it.

DISCUSSION POINT

Discuss the following idea as a class.

Would you choose to learn a new language to speak to new people? What would be easier for you? Would the benefit of being able to talk to all the other people who use that language be useful? Discuss the advantages and disadvantages of both learning the new language, and not learning the new language.

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DID YOU KNOW?

In a survey in 2021, 75% of US adults in the 65+ age group said they used the internet compared with 99% in the 18–29 age group.

Real-life example

Retirement villages (places with houses or apartments intended for older people) are built using several different technologies to show how the technology can benefit their lives. Sensors can monitor a patient's health and doctors can take advantage of high-speed broadband to make video-conference calls to check in and look after older patients. This helps people who find travelling to a clinic more difficult.

Libraries and community centres may offer short IT sessions to help people get started with computers, tablets and using the internet (see Figure 6.6).



Figure 6.6: Technology can be accessed by different age groups.

People living in cities and rural areas

The place where a person lives affects their access to technology. One aspect of this is whether they live in a city or a small town or village.

One of the main services affected by location is access to the internet. Access to high-speed and reliable internet connections are more often found in cities than small villages and towns (see Figure 6.7). This is because the **infrastructure** for the availability of high-speed internet connections is far more developed in city locations.



Figure 6.7: Providing high-speed internet to places in remote locations can be a major challenge.

Because many people live in a city there is a greater demand for high-speed internet connections. There are also many hundreds of people living close to each other. Connecting people living in cities to the internet takes less resource (for example, you could connect 300 people to the internet with 100 m of cable). This means that it is cheaper for internet providers to deliver services in cities.

Some people live in rural areas, many kilometres from larger towns and cities. There may be only 300 people in a village. A city may have many hundreds of thousands of people. Providing a connection over a large distance for a small number of people is not very profitable. So, internet providers may try to avoid investing in high-speed internet in rural areas.

People who do not have high-speed internet access may struggle to access the services available on the internet, which may include online shopping and services offered by some online television broadcasting companies. This would happen if they cannot stream films or TV reliably or consistently and could result in a person feeling isolated.

Questions

- 1
 - a How could a lack of high-speed internet access affect a person's ability to use online learning platforms?
 - b How could this affect the digital divide between those living in a city and those living in a rural location?
- 2
 - a How could a lack of mobile phone internet services affect a person's ability to communicate with others?
 - b What kind of services might they not be able to use?
 - c How might this lack of ability to communicate in certain ways make them feel?

The issue of a lack of access to high-speed internet can extend to mobile telephone internet services. In a rural location, a person may find that they have limited connection to mobile networks, or may not get any connection at all.

Real-life example

Sofia lives in a rural area of Spain. Access to high-speed internet in her village is very limited. She is a furniture designer who works from home. Sofia currently creates her designs and sends them to clients using the internet. She wants to make video calls to her clients to have meetings (see Figure 6.8).

However, Sofia's internet connection does not allow her to use video-conferencing technologies effectively. Sofia's internet connection puts her at a disadvantage compared with furniture designers who have access to high-speed internet connections.



Figure 6.8: Video-conferencing technologies help rural businesses to connect with clients.

Reducing the effects of the digital divide in rural areas

To reduce the effects of the digital divide in rural areas, there are two main issues:

- the access people have to digital devices and technology
- how they connect devices to the internet. This could be via the internet or by mobile networks.

Building infrastructure for high-speed internet or mobile networks can be very challenging. For example, a village could be hundreds of kilometres from the nearest city, or a village may be up in the mountains, or in a remote inaccessible area. One way to provide internet access to remote locations is to use satellite technologies. This reduces the amount of infrastructure needed, while still providing access to the internet.

Real-life example

BharatNet is an initiative of the government of India that focuses on developing infrastructure in rural India to provide affordable internet. They are planning to do this through increasing the infrastructure based on fibre optic cabling (see Figure 6.9). They are also planning to deliver access through satellites in approximately 5300 locations.



Figure 6.9: Building digital infrastructure.

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ACTIVITY 6.03

Imagine you are a government technology representative for your country. You have been tasked with improving internet access in rural areas of your country.

Work with a partner and create a presentation that highlights your plan for improving internet access in rural areas of your country.

Your presentation should also include the advantages that such access will provide.

People with differing levels of education

Using technology can be complex. Different types of technology have different interfaces and are used in different ways. Without a certain level of education in how to use technology, it can often seem challenging.

DID YOU KNOW?

In the UK, 6.3% of adults said they had never used the internet in 2020. That's an estimated 3.4 million people.

Many businesses have online systems to allow people to buy products, book tickets or use services. This is very useful for people who are educated in using technology and are confident in using the systems. But people who are not educated in using these systems may find it frustrating and miss out on opportunities.

There are many different reasons why people are not technologically educated. For example, they may not have access to technology and therefore do not have experience in using it. Some people simply do not have an interest in using technology. Some people

may struggle to keep learning about all the new developments in technology and so ‘give up’.

UNPLUGGED ACTIVITY 6.04

Reflect on a time when you needed to increase your digital literacy, for example, to learn a new technology or a new way of using technology that you were already familiar with.

- What concerns did you have about learning the new skill?
- Did you use any learning techniques that you already knew?
- How did you apply your current level of digital literacy to help develop the new skill?
- What was the outcome of learning this new skill?
- How would you change your approach if you had to learn a new skill again?

Questions

- 3 Think of two advantages that people who are educated in using technology may have over those who are not educated in using technology.

Real-life example

A person wants to buy tickets to see their favourite band. Tickets are only available online. The person has never used an online ticket booking system before and feels very nervous about doing so. They usually call the concert location and purchase tickets over the telephone, but this option is not available.

When the person tries to use the online system to buy the tickets, they are confused and do not know how to operate the system. By the time they work out what to do, the tickets have all sold out. The person now cannot buy the tickets to see the band because they did not have the education in technology and experience needed to navigate the ticket booking system.

Reducing the digital divide by improving digital literacy

One effective way of reducing the digital divide is through digital literacy teaching by providing access to technology courses in local communities. Digital literacy courses for adults are often offered by colleges, schools and libraries. Courses can be offered for free, or at a low cost. This allows people to improve their knowledge of how to use technology. Courses allow people to learn with other people of the same skill level. This helps people feel more comfortable about learning new skills. Local face-to-face courses have a teacher or a leader to help them with problems. People often find face-to-face courses more useful. This is because they may need to learn about skills to access online learning courses. People can then use the skills they learn to use technology more confidently in the future.

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Real-life example

A country offered a 12-week course to its citizens for learning skills in everyday software. The course was provided by a virtual university. The course provided basic knowledge on using software and how to use the internet for communication (see Figure 6.10).

This education provided people with a basic education in the use of technology. It allowed them to feel more comfortable and confident in their skills. The country believed this helped people to be more competitive in the digital world.



Figure 6.10: Virtual education can provide essential digital literacy skills.

People in different socio-economic groups

The digital divide between different socio-economic groups often revolves around cost. Richer people can afford higher-speed internet connections and expensive advancements in technology. This improves their standard of living because they have better access to:

- information and learning
- entertainment
- online services such as shopping
- worldwide communication.

Lack of access to information tends to have a greater impact on people with lower levels of education. The lack of access to services like free education can create a greater digital divide between the rich and the poor. Ensuring access for all to education and information is an important way of improving the socio-economic circumstances.

Questions

- 4 How could the ability to access free online learning platforms support people in poorer communities?
- 5 How could access to high-speed internet connections for entertainment purposes improve the lifestyle of a person?

Real-life example

Taliya is a film studies student. She has high-quality recording equipment, a 4K monitor and a high-spec computer system at home. She really enjoys recording scenes for her film projects and editing them on her computer system at home.

Jaz is also a film studies student. He is becoming increasingly frustrated as he only has a low-spec mobile phone to film and edit all his film projects. He knows that he could produce much better work, and enjoy his studies far more, if he had access to a better computer system to edit his films with.

Reducing the digital divide in different socio-economic groups

Many countries try to provide technology at lower cost (or even for free) to those in poorer communities. This helps close the digital divide by giving access to the advantages that the technologies can provide.

Large organisations, such as Facebook and Google, are working with internet service providers. They help to build infrastructure to bring internet access to areas or countries that cannot afford it. For example, they help to reduce the costs of installing Wi-Fi access points. Organisations collaborate in this way because it means more people have access to the internet. This means that more people may use the services that these organisations provide.

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People with accessibility barriers

The digital divide can affect people who may have specific challenges in accessing a particular technology. For example, someone living with vision loss will not be able to use a digital display effectively (see Figure 6.11).



Figure 6.11: A person with a spinal cord injury overcomes accessibility barriers by using their thumb to access and use their laptop.

Support for people with accessibility barriers is important when digital technology is developed. Many digital devices have some support included. For example, there are

screen readers that convert the text on the screen into sound. Effectively, a person can hear what is on the screen as the digital device reads it out to them. Voice input/commands can be used to help those who may not be able to operate keyboards or touchscreens effectively.

People in more and less industrially developed nations

The parts of the world that have greater access to modern technology are the USA, Europe and northern Asia. These areas mostly consist of more industrially developed countries. Areas where technology access is more restricted include parts of Africa, India and southern Asia. These are areas that are less industrially developed. Access to technology may be less in these areas because the countries may need to use their financial resources to provide other important services. They may not have the money to spend on the development of their technology services.

The costs for countries to build the infrastructure for technology use are very high. However, not having access to technology and the internet could put them at both a competitive and economic disadvantage. Lack of technology may limit trade opportunities and therefore they may fall behind financially when compared to more industrially developed countries. It may also affect the level of education that can be provided to the country's citizens. They may not be able to access the learning and information available to them on the internet. This can have an impact on both the education and the skill level of people in those countries and prevent them from competing on an international level.

DISCUSSION POINT

Business and education are two areas where the digital divide can restrict the industrial development of a country. Work with a partner. Think of one other area that could be affected and discuss what impact this could have on the country.

ACTIVITY 6.05

How aware are you of the technological infrastructure in your country?

- 1 Make a list (without a computer) of the technology facilities that you think the government of your country provides to its citizens.
- 2 Use the internet to research the list of facilities the government provides.
- 3 How many did you get right? Were there any you are surprised by?
- 4 Do further research using the internet to find out any future plans your country has to improve its technological infrastructure.

Real-life example

A village cooperative in Kenya makes hand-beaded necklaces that it sells to local people and tourists. The members would like to expand the business and sell their necklaces to a wider market. This would increase the cooperative's income and benefit everyone in the village (see Figure 6.12). The cooperative sets up an online business, but internet access to

the village is poor and it is unable to run the website effectively. The lack of infrastructure has put the cooperative at a competitive disadvantage.

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Figure 6.12: Internet access allows people across the world to see a business's products and place orders for them online.

The village cooperative wants to improve access to the internet in their village. Developing internet access is expensive and the cooperative may not be able to pay for this without help. Local communities often create campaigns to help raise awareness of the local facilities, and what they want in the future. The campaign will be sent to the local government, or maybe even the national government. They would create a campaign to ask for better services, and show how these services could impact their community. It is important to remember that the members of the cooperative may have a low level of technology awareness. They may not know about the best digital technology or how to use it. Part of the campaign may involve a consultation with IT specialists from the government. This would allow the government to show what technology is available, and for the members of the cooperative to see how they can use the technology. A plan can then be put in place that balances the technology and costs against the impact on the cooperative.

Reducing the digital divide in less industrially developed areas

The internet has become essential to global infrastructures. Those with limited access to the internet may not be able to access all the benefits of the internet. Satellite technology may assist remote areas to access the internet. Large companies plan to use satellite technologies that allow such areas better access to the internet.

DID YOU KNOW?

A company called Starlink is aiming to provide high-speed internet connections to the world using satellites in space. Starlink launches satellites into space that beam internet signals to Earth. This allows internet access to many places where cables and

other wireless connections cannot reach. This means that millions of people could soon be able to access the internet without the need for local infrastructure. The challenge is that satellite technology is expensive to run. This means that the cost to access the internet by satellite is quite high when compared to using ground-based access.

Governments in many less industrially developed regions try to promote the business and products that they could offer if they had better infrastructure and high-speed access to the internet. They are keen for more industrially developed countries to invest and help them to build the necessary infrastructure. These coordinated efforts can benefit the world on a global scale.

Real-life example

Microsoft has an initiative called the Microsoft Airband Initiative. The aim of the initiative is to close the global digital divide by working with its partners to provide internet access to millions of people worldwide whose communities have limited or no internet access. The initiative uses a range of wired and wireless technologies. One part of the initiative uses Television White Spaces (TVWS) technology. TVWS is fixed wireless technology that uses radio frequencies which currently broadcast television. Internet access can be provided using these radio frequencies that are often unused and unlicensed. This technology is giving internet access to people in countries where the cost of adding ground-based infrastructure is either very expensive or would take a long time to build.